Supplementary material for "Lightweight Large Language Model-Based Semantic Dispatch Framework for Electric Vehicle Virtual Power Plants"

I. EV-Level Flexibility Envelope Construction

For charging station s, the EV-level upper and lower flexibility boundaries used in the main paper are constructed as

$$p^{+}_{s,i,\kappa} = \begin{cases} p^{\max}_{s,i}, & a_{s,i} \le \kappa \le d_{s,i}, \\ 0, & \kappa < a_{s,i} \text{ or } \kappa > d_{s,i}, \end{cases} \tag{1a}$$

$$p^{-}_{s,i,\kappa} = \begin{cases} -v_{s,i} p^{\max}_{s,i}, & a_{s,i} \le \kappa < d_{s,i}, \\ 0, & \kappa < a_{s,i} \text{ or } \kappa \ge d_{s,i}, \end{cases} \tag{1b}$$

$$e^{+}_{s,i,\kappa} = \begin{cases} 0, & \kappa \le a_{s,i}, \\ \min\left\{ \begin{matrix} e^{+}_{s,i,\kappa-1} + \Delta T^{\kappa} \eta_{s,i} p^{+}_{s,i,\kappa}, \\ e^{\text{tar}}_{s,i} \end{matrix} \right\}, & a_{s,i} < \kappa < d_{s,i}, \\ e^{\text{tar}}_{s,i}, & \kappa \ge d_{s,i}, \end{cases} \tag{1c}$$

$$e^{-}_{s,i,\kappa} = \begin{cases} 0, & \kappa \le a_{s,i}, \\ \max\left\{ \begin{matrix} e^{-}_{s,i,\kappa-1} + \Delta T^{\kappa} \dfrac{p^{-}_{s,i,\kappa}}{\eta_{s,i}}, \\ 0 \end{matrix} \right\}, & a_{s,i} < \kappa \le \kappa^{\text{L}}_{s,i}, \\ \min\left\{ \begin{matrix} e^{-}_{s,i,\kappa-1} + \Delta T^{\kappa} \eta_{s,i} p^{+}_{s,i,\kappa}, \\ e^{\text{tar}}_{s,i} \end{matrix} \right\}, & \kappa^{\text{L}}_{s,i} < \kappa < d_{s,i}, \\ e^{\text{tar}}_{s,i}, & \kappa \ge d_{s,i}, \end{cases} \tag{1d}$$

$$\kappa^{\text{L}}_{s,i} = \max\left\{ a_{s,i},\ d_{s,i} - \left\lceil \frac{e^{\text{tar}}_{s,i}}{\Delta T^{\kappa} \eta_{s,i} p^{\max}_{s,i}} \right\rceil \right\}. \tag{1e}$$

where $\kappa^{\text{L}}_{s,i}$ is the latest interval at which EV (s,i) can stay on the lower energy envelope while still reaching the departure target.

II. Charging-Station Feasible Envelope Aggregation

Based on the EV-level flexibility boundaries in (1), the station-level feasible envelope used as $\mathcal{F}^{\text{CS}}_{s,\kappa}$ in the main paper is constructed as

$$P^{+}_{s,\kappa} = \sum_{i \in \mathcal{N}_s} p^{+}_{s,i,\kappa}, \qquad P^{-}_{s,\kappa} = \sum_{i \in \mathcal{N}_s} p^{-}_{s,i,\kappa}, \tag{2a}$$

$$E^{+}_{s,\kappa} = \sum_{i \in \mathcal{N}_s} e^{+}_{s,i,\kappa}, \qquad E^{-}_{s,\kappa} = \sum_{i \in \mathcal{N}_s} e^{-}_{s,i,\kappa}, \tag{2b}$$

$$E^{\text{CS}}_{s,\kappa} = \begin{cases} E^{\text{CS}}_{s,\kappa-1} + \Delta T^{\kappa} \eta^{\text{CS}}_{s} P^{\text{CS}}_{s,\kappa}, & P^{\text{CS}}_{s,\kappa} \ge 0, \\ E^{\text{CS}}_{s,\kappa-1} + \Delta T^{\kappa} \dfrac{P^{\text{CS}}_{s,\kappa}}{\eta^{\text{CS}}_{s}}, & P^{\text{CS}}_{s,\kappa} < 0, \end{cases} \tag{2c}$$

$$P^{-}_{s,\kappa} \le P^{\text{CS}}_{s,\kappa} \le P^{+}_{s,\kappa}, \tag{2d}$$

$$E^{-}_{s,\kappa} \le E^{\text{CS}}_{s,\kappa} \le E^{+}_{s,\kappa}. \tag{2e}$$

where $\mathcal{N}_s$ is the set of EVs connected to charging station s; $P^{+}_{s,\kappa}$, $P^{-}_{s,\kappa}$, $E^{+}_{s,\kappa}$, and $E^{-}_{s,\kappa}$ are the station-level upper and lower power and energy envelopes; $P^{\text{CS}}_{s,\kappa}$ and $E^{\text{CS}}_{s,\kappa}$ are the aggregated station power and energy state; and η^{CS}_{s} is the station-level charging/discharging efficiency. Positive $P^{\text{CS}}_{s,\kappa}$ denotes net charging, and negative denotes net discharging. In the intra-day model, $\mathcal{F}^{\text{CS}}_{s,\tau}(\zeta_{s,\tau})$ keeps the station power and energy update constraints in (2) but relaxes the lower energy-envelope bound as

$$E^{\text{CS}}_{s,\tau} + \zeta_{s,\tau} \ge E^{-}_{s,\tau}, \qquad E^{\text{CS}}_{s,\tau} \le E^{+}_{s,\tau}, \qquad \zeta_{s,\tau} \ge 0. \tag{3}$$

III. Intra-Day Objective Cost Definitions

The compact intra-day objective in the main paper consists of three cost terms:

$$C_{\text{ene}}^{\text{RT}} = \sum_{\tau \in \mathcal{T}^{\text{RT}}} \hat{\lambda}_{\tau}^{\text{RT}} \left(P_{\tau}^{\text{VPP,RT}} - P_{\tau}^{\text{VPP,DA}} \right) \Delta T^{\text{RT}}, \tag{4a}$$

$$C_{\text{op}}^{\text{RT}} = \sum_{\tau \in \mathcal{T}^{\text{RT}}} \sum_{s \in \mathcal{S}} c^{\text{op}} \left| P_{s,\tau}^{\text{CS}} \right| \Delta T^{\text{RT}}, \tag{4b}$$

$$C_{\text{env}}^{\text{RT}} = c^{\text{env}} \sum_{\tau \in \mathcal{T}^{\text{RT}}} \sum_{s \in \mathcal{S}} \zeta_{s,\tau}. \tag{4c}$$

where $C_{\text{ene}}^{\text{RT}}$ is the intra-day energy-settlement cost, $C_{\text{op}}^{\text{RT}}$ represents the CS operation cost, and $C_{\text{env}}^{\text{RT}}$ penalizes relaxation of the lower energy envelope.

IV. EV-Level Disaggregation Model

After the station-level dispatch power is obtained, the EV-level disaggregation model allocates the station reference to individual EV charging and discharging trajectories. For CS s, let $P_{s,\tau}^{\text{ref}}$ denote the station-level reference power. The disaggregation problem is formulated as:

$$\min \sum_{\tau \in \mathcal{T}^{\text{RT}}} \left(P_{s,\tau}^{\text{CS}} - P_{s,\tau}^{\text{ref}} \right)^2 + c^{\text{sf}} \sum_{i \in \mathcal{N}_s} \sigma_{s,i} \tag{5a}$$

$$P_{s,\tau}^{\text{CS}} = \sum_{i \in \mathcal{N}_s} p_{s,i,\tau}^{\text{ev}} \tag{5b}$$

$$p_{s,i,\tau}^{\text{ev}} = p_{s,i,\tau}^{\text{ch}} - p_{s,i,\tau}^{\text{dis}} \tag{5c}$$

$$e_{s,i,\tau+1} = e_{s,i,\tau} + \Delta T^{\text{RT}} \eta_{s,i} p_{s,i,\tau}^{\text{ch}} - \Delta T^{\text{RT}} \frac{p_{s,i,\tau}^{\text{dis}}}{\eta_{s,i}} \tag{5d}$$

$$0 \leq p_{s,i,\tau}^{\text{ch}} \leq p_{s,i}^{\max} \delta_{s,i,\tau}^{\text{ch}}, \qquad 0 \leq p_{s,i,\tau}^{\text{dis}} \leq v_{s,i} p_{s,i}^{\max} \delta_{s,i,\tau}^{\text{dis}} \tag{5e}$$

$$\delta_{s,i,\tau}^{\text{ch}} + \delta_{s,i,\tau}^{\text{dis}} \leq 1, \qquad \delta_{s,i,\tau}^{\text{ch}}, \delta_{s,i,\tau}^{\text{dis}} \in \{0,1\} \tag{5f}$$

$$\begin{cases} e_{s,i,\tau} = 0, & \tau \leq a_{s,i} \\ 0 \leq e_{s,i,\tau} \leq e_{s,i}^{\text{tar}}, & a_{s,i} < \tau < d_{s,i} \\ e_{s,i,\tau} + \sigma_{s,i} = e_{s,i}^{\text{tar}}, & \tau \geq d_{s,i} \end{cases}, \qquad \sigma_{s,i} \geq 0 \tag{5g}$$

where $\sigma_{s,i}$ is the departure-energy shortfall of EV (s,i), and c^{sf} is the corresponding penalty coefficient. The first objective term tracks the station-level reference power, while the second term penalizes unmet departure energy.

V. Key LLM Hyperparameter Settings

This supplementary file reports the main LLM hyperparameters used in the semantic parsing experiments. Only the settings that materially affect parser training and inference are included here.

A. Supervised Fine-Tuning

The first-stage supervised fine-tuning adapts the lightweight model to the dispatch-oriented semantic event schema and the required structured JSON output protocol.

TABLE I
Main Hyperparameters for LoRA-SFT

Item	Setting
Base model	Qwen2.5-0.5B-Instruct
Training samples	600
Fine-tuning method	LoRA-SFT
LoRA rank / alpha / dropout	16 / 32 / 0
Maximum sequence length	4096 tokens
Learning rate	5.0×10^{-5}
Scheduler	Cosine
Optimizer	AdamW
Epochs	3
Per-device batch size	2
Gradient accumulation steps	8
Numerical precision	bfloat16

B. Reinforcement Fine-Tuning

The second-stage reinforcement fine-tuning is initialized from the LoRA-SFT model. The compared GRPO-family methods use the same policy-training hyperparameters, while differing in reward construction.

TABLE II
MAIN HYPERPARAMETERS FOR GRPO-FAMILY FINE-TUNING

Item	Setting
Initialization	LoRA-SFT model
Training samples	600
Fine-tuning method	LoRA-based GRPO-family update
LoRA rank / alpha / dropout	16 / 32 / 0.1
KL coefficient	0.04
Sampling temperature during training	0.9
Learning rate	3.0×10^{-5}
Scheduler	Cosine
Optimizer	AdamW
Weight decay	0.1
Warmup ratio	0.1
Epochs	15
Per-device batch size	2
Gradient accumulation steps	4
Number of sampled generations	4
Maximum prompt length	4096 tokens
Maximum completion length	1024 tokens
Maximum gradient norm	0.1
Numerical precision	bfloat16

TABLE III
REWARD-SPECIFIC SETTINGS

Method	Reward setting	Buffer size	Update rate	Bonus
GRPO	Fixed parsing reward and format-boundary reward	–	–	–
LoRA-GRPO	Fixed parsing reward and format-boundary reward	–	–	–
LoRA-SGRPO	Static error-aware reward over four parsing error types	300	0	0.25
LoRA-AGRPO	Adaptive error-aware reward over four parsing error types	300	0.02	0.25

C. Parser Inference

The complete semantic dispatch scheme uses the LoRA-AGRPO parser for online conversion from natural-language requests to structured semantic events.

TABLE IV
MAIN HYPERPARAMETERS FOR PARSER INFERENCE

Item	Setting
Parser model	Qwen2.5-0.5B LoRA-AGRPO parser
Test samples	300
Maximum new tokens	2048
Top-p	0.7
Temperature	0.95
Device	Single GPU
Output protocol	Boundary-delimited JSON